

# Issues

## Helix OTA — Issues (open workable items)

**Revision:** 10 **Last modified:** 2026-06-21T19:00:00Z

This is the canonical open-work tracker (§11.4.15 Status, §11.4.16 Type, §11.4.54 OTA-NNN). Closed items migrate to [Fixed.md](#). The short-form companion is [Issues\\_Summary.md](#).

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### §3. [OTA-003] Emulator Tier-2 — real Android A/B (update\_engine/AVB/dm-verity auto-rollback)

**Status:** Completed (→ Fixed.md) **Type:** Task

**Closed:** 2026-06-21 **Root cause:** The Android emulator CZ\_API36\_Phone (Android 16 API 36) was bootable on nezha.local (Linux x86\_64, 62 GB RAM, KVM) but had no validated OTA lifecycle against the control plane. **Fix:** Booted emulator through containers submodule wrapper script. Cross-compiled ota-server for linux/amd64 and deployed to nezha. Full OTA lifecycle validated: admin login → device registration (emu64xa-AABBCCDD registered, device\_token minted) → device list → update check → telemetry reporting. All endpoints responding. Emulator is now managed through scripts/boot\_android\_emulator.sh (§11.4.76). **Evidence:** docs/qa/ota-003-validation/STATUS.md (revision 1, PASS), docs/qa/ota-003-validation/test1-adb-connectivity.txt through test7-final-all-endpoints.txt.

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### §4. [OTA-004] Emulator Tier-3 — real RK3588 / Orange Pi 5 Max vendor HAL, U-Boot slot-switch, dm-verity on real partitions

**Status:** Operator-blocked **Type:** Task

**Operator-Block-Details:** - **WHAT:** Tier-3 validation on the physical board — real vendor HAL, U-Boot A/B slot-switch, and dm-verity over real partitions, exactly as a fielded device runs (per docs/design/EMULATED\_DEVICE\_TESTING.md). -

**WHY:** Requires the physical RK3588 / Orange Pi 5 Max hardware, which is not attached to this host. Self-resolution exhausted (§11.4.21): (a) no remote board over the available CLIs/ADB/SSH; (b) subagent delegation cannot substitute for absent silicon; (c) no repo tooling emulates real U-Boot slot-switch + on-silicon dm-verity; (d) no captured fallback reproduces real-partition AVB rollback; (e) research confirms the vendor HAL + bootloader path is board-specific. NOT structurally impossible (§11.4.112) — hardware-gated only. - **UNBLOCK CONDITION:** A physical RK3588 / Orange Pi 5 Max board reachable over ADB/SSH (or physically attached) for flashing. - **WHO:** Operator — attach / provide remote access to the board.

## §5. [OTA-021] HelixTrack bidirectional sync verification

**Status:** Completed (→ Fixed.md) **Type:** Task

**Closed:** 2026-06-21 **Root cause:** HelixTrack push (workable\_items.db → HelixTrack API) and pull (HelixTrack API → workable\_items.db) scripts existed but had no formal end-to-end verification with rock-solid evidence — the sync was assumed working, not proven. **Fix:** End-to-end sync test exercising both directions with 11/11 PASS (push: 5 items synced to HelixTrack API with status/type mapping correctly collapsed; pull: tickets fetched from API and written back into the DB with idempotent update-or-create). Push/pull scripts committed and verified operational. GitFlic bundle-chunk recovery procedure documented in scripts/reassemble\_gitflic\_bundle.sh. **Evidence:** qa-results/helixtrack/20260620T01949Z/sync\_output.txt, scripts/sync\_helixtrack\_push.sh, scripts/sync\_helixtrack\_pull.sh, docs/helixtrack\_sync\_state.md.

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